

Imad EL BADISY

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PROFILE

Research Engineer in Biostatistics and Machine Learning, PhD, working at the interface of computational biostatistics and machine learning. Moroccan and French national, husband and father. I hold a PhD from Aix-Marseille University (SESSTIM laboratory), completed under the supervision of Pr Roch GIORGI, on missing data and machine learning imputation methods for survival analysis. My work focuses on building methodological tools for health data science supporting biomedical research projects. I develop open-source R packages for survival analysis, missing data, and interpretable machine learning, and contribute as a consultant and trainer in biostatistics and health data science.

EXPERIENCE

Mohammed VI Center for Research and Innovation

Rabat, Morocco

(CM6RI)

Head of AI and Data Science Service

Nov. 2021 – Present

- Leading the AI and Data Science service
- Conducting methodological and applied research in health (oncology, fertility, other biomedical domains)

The GRAPH Network

Switzerland / Remote

Data Science Education Officer

Sep. 2023 – Dec. 2023

- Designed and delivered training programs in data science for global health
- Developed educational material and course content for data science learners

Institut Mines-Télécom, IMT Atlantique

Brest, France

Research Engineer in Applied Statistics

Sep. 2019 – Sep. 2021

- Conducted statistical analyses of national health data (SNDS-based analyses)
- Implemented automated and reproducible data workflows

INSERM

Clermont-Ferrand, France

Biostatistician

Jan. 2019 – Jun. 2019

- Conducted cost-effectiveness evaluation of a clinical protocol on ketamine for chronic pain

Université de Sherbrooke

Sherbrooke, Quebec, Canada

Health Economist Intern

Feb. 2017 – Sep. 2017

- Conducted a systematic review and meta-analysis on health utilities in cancer patients

EDUCATION

Aix-Marseille University, SESSTIM

Marseille, France

PhD in Biostatistics

2021 – 2025

Thesis: Missing Data and Machine Learning Methods for Survival Analysis · Supervisor: Pr Roch Giorgi

Aix-Marseille University, SESSTIM

Marseille, France

Master 2 in Quantitative Methods for Health Research

2018 – 2019

Université Clermont Auvergne

Clermont-Ferrand, France

Master 1 in Applied Mathematics and Statistics

2017 – 2018

Université Clermont Auvergne, CERDI

Clermont-Ferrand, France

Master 2 in Health Economics

2017

SOFTWARE (R PACKAGES)

survalis — Interpretable Survival Machine Learning

CRAN · v0.7.1

Framework

- 19 Survival learners, 7 evaluation metrics, 8 interpretability methods · Maintainer

funcml — Functional Machine Learning Framework

CRAN

- Unified framework for functional machine learning in R · Maintainer

survdnn — Deep Neural Networks for Survival Analysis (R torch)	CRAN
• Cox, AFT, and discrete-time neural survival models · Maintainer	
tvrmst — Time-Varying RMST from Survival Matrices	CRAN
• Dynamic RMST summaries for individualized survival curves · Maintainer	
unsurv — Unsupervised Clustering of Individualized Survival Curves	CRAN
• Phenotyping patients from predicted survival curves · Maintainer	
mcstatsim — Monte Carlo Statistical Simulation Tools	CRAN
• Functional approach to monte carlo simulation studies in R · Maintainer	
missCforest — Ensemble Conditional Trees for Missing Data Imputation	CRAN
• Missing data imputation using conditional forest ensembles · Maintainer	

SELECTED PUBLICATIONS

El Badisy I., Assarag B., Belrhiti Z. (2026). Interpretable clinical decision support in high-risk pregnancy: A scoping review. *BMC Pregnancy and Childbirth*.

Kadi C., Ahmadi N., Houdou A., **El Badisy I.**, et al. (2025). Differentiating latent from active tuberculosis through activation phenotypes and chemokine markers. *Biomarker Insights*, 20, 11772719241312776.

El Badisy I., Graffeo N., Khalis M., Giorgi R. (2024). Multi-metric comparison of ML imputation methods: application to breast cancer survival. *BMC Medical Research Methodology*, 24(1):191.

El Badisy I., BenBrahim Z., Khalis M., et al. (2024). Risk factors for colorectal cancer survival in Morocco: interpretable ML approach. *Scientific Reports*, 14(1):3556.

Houdou A., **El Badisy I.**, Khomsi K., et al. (2024). Interpretable ML for forecasting air pollution: a systematic review. *Aerosol and Air Quality Research*, 24(1):230151.

Zbiri S., Belghiti Alaoui A., **El Badisy I.**, et al. (2024). Private hospitals in LMICs: a typology using cluster methods — Morocco. *BMC Health Services Research*, 24(1):1231.

Blom I.M., Eissa M., Mattijsen J.C., **El Badisy I.**, et al. (2023). Greenhouse gas mitigation interventions for health-care systems. *Bulletin of the World Health Organization*, 102(3):159.

Khalis M., Toure A.B., **El Badisy I.**, et al. (2022). Meteorological parameters and COVID-19 in Casablanca, Morocco. *Int. J. Environmental Research and Public Health*, 19(9):4989.

TEACHING

AI for Public Health — Aix-Marseille University, SESSTIM	2022 – Present
• Machine learning methods and NLP for public health · Master's students	
Introduction to Biostatistics — University Mohammed VI of Health Sciences (UM6SS)	2021 – Present
• Descriptive and inferential statistics, statistical reasoning for health sciences · 100+ hours/year	
Quantitative Epidemiology and Machine Learning — Institut Mines-Télécom, IMT Atlantique	2020 – Present
• ML fundamentals, quantitative epidemiology, supervised and unsupervised learning · 21+ hours/year	

Outside working hours: {**Methods in Health Data Science with R**} — an independent online course pathway covering applied biostatistics, survival analysis, machine learning, and meta-analysis with R. mhdsr.com

SELECTED TALKS

Comparison of missing data imputation methods in Cox regression with time-varying effects <i>ISCB 2023, Rome</i> · El Badisy I. et al.	2023
Missing data imputation using a meta-algorithm metaCART: simulation under MCAR	2023

SKILLS

Methods: Biostatistics · Survival analysis · Missing data · Meta-analysis · Interpretable machine learning

Tools: R package development · Python · Quarto/LaTeX · Git/GitHub · Slurm/HPC

Languages: Arabic (native) · French (fluent) · English (fluent)

Nationality: Moroccan · French